

Rex Tin Long Fung

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Website: <https://rextlfung.github.io>

EDUCATION

University of Michigan, Ann Arbor, MI Sep 2022 - Present
PhD Candidate specializing in MRI sequence programming and iterative reconstruction methods, with applications to accelerating functional MRI.

University of Michigan, Ann Arbor, MI Sep 2022 - Aug 2024
Program: M.S., Biomedical Engineering (Imaging and Ultrasonics) **GPA:** 3.65/4.00
Relevant Coursework: Linear Algebra, Optimization, Probability, Machine Learning, Image Processing, Medical Imaging Systems

University of Michigan, Ann Arbor, MI Sep 2018 - May 2022
Program: B.S., Neuroscience major, Electrical Engineering minor **GPA:** 3.78/4.00
Relevant Coursework: Digital Signal Processing, Electrical Biophysics, Embedded Control Systems, Sensing and ML for Neural Interfaces

RESEARCH EXPERIENCE

Functional MRI Lab, University of Michigan Sep 2022 - Present
Graduate Student Research Assistant *Advisors: Dr. Jon-Fredrik Nielsen, Dr. Douglas Noll*

- Accelerated fMRI data acquisition by $6\times$ by developing a novel 3D-EPI sequence with Poisson-disc random sampling, increasing resolution for functional neuroimaging studies.
- Simultaneously reconstructed 100 GB+ fMRI datasets on HPCs using a custom 4D iterative optimization algorithm written in Julia, which models low-rank signal structures on multiple scales to improve accuracy of functional activation maps.
- Standardized cross-vendor research for 7+ international institutions (e.g., Harvard, Freiburg) by developing open-source Pulseseq sequences, facilitating seamless, hardware-agnostic protocol sharing.
- Co-authored 1 journal article and 3 conference posters at top-tier venues (e.g. IEEE, ISMRM), translating complex sequence design and reconstruction findings for the global imaging community.

Bo Duan Lab, University of Michigan Jul 2020 - May 2022
Undergraduate Research Assistant *Advisor: Dr. Bo Duan*

- Reduced manual data processing time by 90% by using OpenCV to automate behavioral tracking in 30+ hours of footage and cell-counting for 200+ microscopy images.

TEACHING EXPERIENCE

BME/ECE 516 Medical Imaging Systems, University of Michigan Sep 2024 - Dec 2024

- Mentored 20-30 students to learn Medical Imaging Systems e.g. MRI, ultrasound, by guiding them through homework problems during weekly office hours and via Piazza, an online Q&A forum.
- Streamlined course logistics for 15 weeks by releasing and grading homework assignments + final exam on Canvas, an online site for the class.

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SKILLS

Math: Inverse Problems, Optimization, Transform Domains, Machine Learning

Programming: Julia, Python, MATLAB, GPU programming, Jupyter Notebooks, LaTeX, Git/GitHub

Science: Experimental Design, Oral Presentation, Academic Writing

Languages: Cantonese, Mandarin Chinese, Japanese

Sports: Volleyball, Boulderling, Badminton, Pickleball, Table Tennis, Basketball, Triathlon

Digital Media Production: Adobe Premiere Pro/Rush, Audacity, Edits

JOURNAL PUBLICATIONS

- [1] R.A. Lobos, X. Wang, **R.T.L. Fung**, Y. He, D. Frey, D. Gupta, Z. Liu, J.A. Fessler, D.C. Noll. "Spatiotemporal Maps for Dynamic MRI Reconstruction.", *IEEE Transactions on Computational Imaging* (2026). [DOI](#).
- [2] H. Lee, C.C. Hor, L.R. Horwitz, A. Xiong, X.Y. Su, D.R. Soden, S. Yang, W. Cai, W. Zhang, C. Li, C. Radcliff, A. Dinh, **T.L.R. Fung**, I. Rovcanin, K.P. Pipe, X.Z.S. Xu, B. Duan. "A dedicated skin-to-brain circuit for cool sensation in mice.", *Nature Communications* (2025). [DOI](#).

CONFERENCE POSTERS

- [1] **R.T.L. Fung**, R.A. Lobos, J.A. Fessler, D.C. Noll, J.-F. Nielsen. "Sub 2 mm resolution fMRI at 3T using randomly undersampled 3D-EPI with locally low-rank + temporally sparse reconstruction" *International Society for Magnetic Resonance in Medicine Annual Meeting*, 2025. Digital poster #3196.
- [2] S. Peltier, M. Egan, **R. Fung**, Q. Chen, M. Zaitsev, J.-F. Nielsen. "Harmonization of an SMS-EPI fMRI protocol using Pulseseq: Sequence implementation and ABCD QC metrics." *Organization for Human Brain Mapping Annual Meeting*, 2024. Digital poster #3804.
- [3] Y. He, **R. Fung**, J.-F. Nielsen. "High-Accuracy Ultra-short Inner-Volume Saturation Pulse for 3D Steady-State Imaging." *International Society for Magnetic Resonance in Medicine Annual Meeting*, 2024. Digital poster #3249.

HONORS AND AWARDS

James B. Angell Scholar, University of Michigan

Mar 2021

Students who achieve an "A" record (all grades of A+, A, or A-) for two or more consecutive terms